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22CH203 ENGINEERING CHEMISTRY-II

UNIT I- & ELECTROCHEMISTRY

LESSON PLAN 1.1

Topic: ***Introduction to Electrochemistry***

Reference: ***Chemistry for Engineering students by Lawrence S Brown and Thomas A Holme (Fourth Edition) Chapter 13***

1. INTRODUCTION

Electrochemistry is a branch of chemistry which deals with the chemical applications of electricity. It primarily deals with the chemical reactions produced by passing electric current through an electrolyte or the production of electric current through chemical reactions.

Some examples of energy transformation

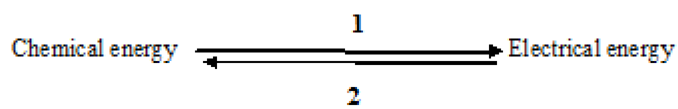
Chemical energy from food is converted to mechanical energy when the food is broken down and absorbed in the muscles. The chemical energy from food can also be converted to thermal energy to keep the body warm.

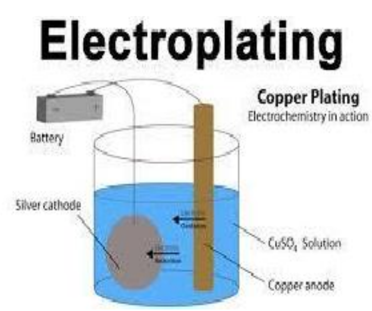
When lightning strikes a tree, electrical energy is converted to thermal energy.

Plants convert light energy to chemical energy through photosynthesis.

Electrical energy can also be converted to sound energy when it is used to power a loudspeaker. Sound energy is converted to electric energy in a microphone.

Waves convert mechanical energy to electrical energy.





1 – Daniel cell; Batteries 2 – Electroplating

When electric current is passed through a conducting aqueous solution (electrolyte), the positively charged ions (cations) move towards the negatively charged electrode known as the cathode, while the negatively charged ions (anions) move towards the positively charged electrode called the anode. Thus the electrical energy is converted into chemical energy. Flow of electricity through electrolytes is due to the migration of ions when there is a potential difference applied between the two electrodes. In short, the flow of electricity in an electrolyte results in chemical reaction at the electrodes.

2 TERMINOLOGY

2.1 Conductors

A substance or material that allows electric current to pass through is called a *conductor. The ability of a material* to conduct electric current is called c*onductance.*

Examples: All metals, graphite, fused salts, aqueous solutions of acids, bases, etc

The conductors are broadly classified into two types as follows

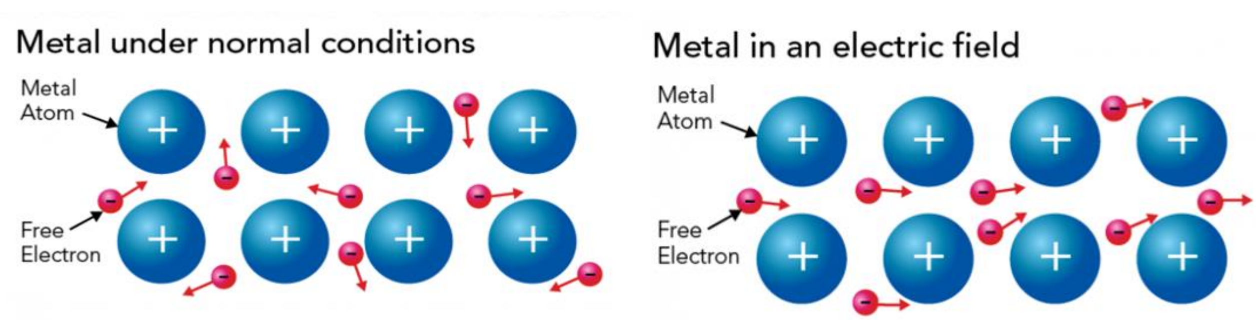
2.1.1 Metallic conductor (or) Electronic conductors

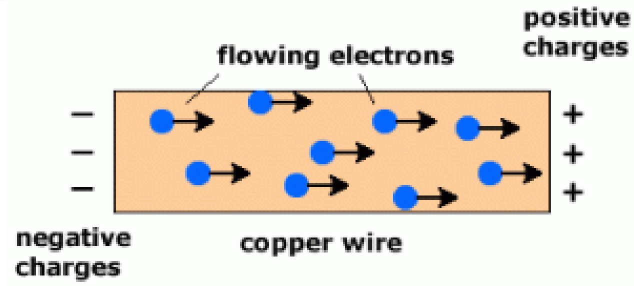
Metallic conductors are solid substances, which conduct electric current due to the movement of electrons from one end to another end. The conduction decreases with increase of temperature.

Examples: All metals, graphite

Flow of electric charges thorough a metallic conductor

In metals, valence electrons are free to move. Free electrons are only loosely attached to their parent atoms. These electrons can move around in all directions. But when they are exposed to a weak electromagnetic field, they all move in one direction. This is the electric current. If lots of electrons pass through a piece of metal in a certain amount of time, the current is strong. If fewer electrons pass, the current is weaker.

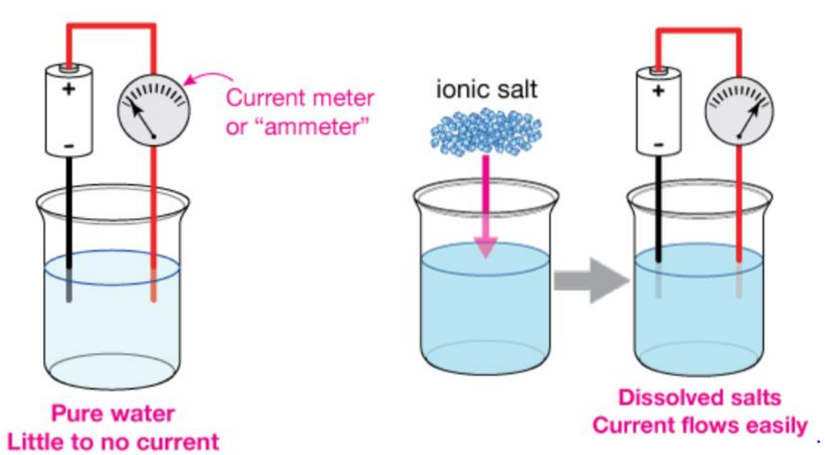


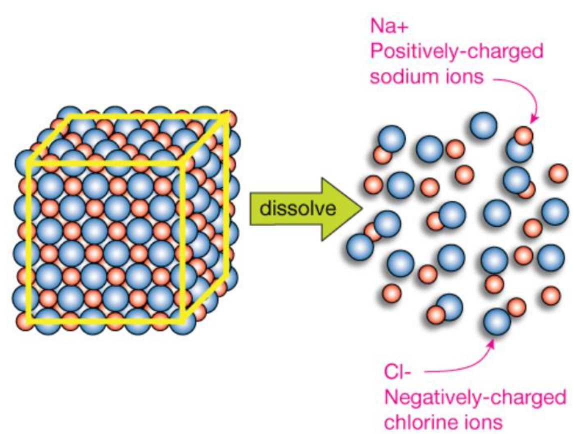


Let us take a conductor, such as a copper metal wire, a flow of negatively charged electrons transports electricity through metallic conductors. The electrons can move from one atom to another because they are free to move. If an electric field is applied, an electric charge will experience a force. If it is free to move, it will contribute to a current. As long as the electrons travel, they will form an electric current. As a result, there will be a current for a very brief time and then no current in the condition under consideration. Free electrons in a conductor are always in a state of random motion with a velocity of the order of 104 m/s and the flow of an electric charge in a conductor without applying a potential difference is zero.

2.1.2 Electrolytic Conductors

Electrolytic conductors conduct electric current due to the movement of ions in solution or in fused state. The conduction increases with increase of temperature. Examples: Acids, bases, electrovalent substances. Pure water is a terrible conductor of electricity. In fact, we often measure the purity of water in laboratories that need high purity in terms of its inability to carry electric current. This can be illustrated by using water to complete a circuit (between the black and red wires) as indicated in first figure:





For pure water, only an exceedingly small current will flow because there's just not enough charged material dissolved in the liquid. On the other hand, if we add some soluble ionic salt, like sodium chloride (NaCl), which breaks apart in water into sodium (Na^+) and chloride (Cl^-) ions (see second figure above), we find that the water can carry a significant current. To varying degrees, natural waters carry current. Sea water, with its high concentration of dissolved salts, is the best of these. Yet even when lightning, which carries a tremendous current, impacts the sea surface, the current rapidly dissipates as it spreads out.

This third figure above shows how highly-ordered crystals of positively-charged sodium ions and negatively-charged chlorine (called chloride) ions of sodium chloride (NaCl, or table salt) break apart when dissolved in water to create ions that can move in the solution and therefore carry current. Ions carry current in many important systems. Animal nerve cells, for example, use a combination of simple ions like Na^+, K^+ and Cl^- to propagate nervous impulses along a chain of neurons at very high speed.

S. No.	Metallic conduction	Electrolytic conduction
1.	It involves the flow of electrons in a conductor.	It involves the movement of ions in a solution.
2.	It does not involve any transfer of matter.	It involves transfer of electrolyte in the form of ions.
3.	Conduction decreases with increase in temperature.	Conduction increases with increase in temperature.
4.	No change in chemical properties of the conductor.	Chemical reactions occur at the two electrodes.

2.2 Electrode

2.2.1 What is an Electrode?

An electrode is a solid electric conductor that carries electric current into non-metallic solids, or liquids, or gases, or plasmas, or vacuums. Electrodes are typically good electric conductors, they are not restricted to metals.

2.2.2 Cathode and Anode

Anode : *Anode is the electrode at which oxidation (loss of electron) occurs.*

Cathode : Cathode is the electrode at which reduction (gain of electron) occurs.

Whether an electrode operates as a cathode or anode depends on the direction the cell is operating in. If a cell is switched from operating galvanically (i.e. output of energy like a battery) to electrolysis (energy is input to the cell) then its cathode will become its anode and vice versa.

2.2.3 Electrode Examples

- In Analytical Chemistry : Examples of typical materials used for electrodes in analytical chemistry are amorphous carbon, gold, and platinum. Glass electrodes are often used in pH measurements; in this application the glass is chemically doped to be selective to hydrogen ions.
- In Batteries: Batteries contain a variety of electrodes, depending on the battery type. Lead-acid batteries are based on lead electrodes. Zinc-carbon batteries are made with zinc and amorphous carbon electrodes.
- Lithium polymer batteries have electrodes made of a solid polymer matrix within which lithium ions can move and act as charge carriers.
- In Electrolysis: Electrical energy can be used to convert salts and ores to metals. Sodium metal is produced by electrolysis using a carbon anode and an iron cathode.

2.3 Electrolyte

Electrolyte is a water soluble substance forming ions in solution and conduct an electric current. The electrolytic conductors are further sub-classified into three types as follows

(a) Strong electrolytes

Strong electrolytes are substances, which ionise completely almost at all dilutions. *Examples: HCl, NaOH, KCl etc.,*

(b) Weak electrolytes

Weak electrolytes are substances, which ionise to a small extent even at high dilutions.*Examples:*CH_3COOH, NH_4OH, $CaCO_3$

(c) Non electrolytes

Non electrolytes are substances, which do not ionise at any dilutions. *Examples:Glucose, Starch, alcohol, petrol*

2.4 Anode Compartment

Anode compartment is the compartment of the cell in which oxidation half-reaction occurs. It contains the anode.

2.5 Cathode Compartment

Cathode compartment is the compartment of the cell in which reduction half reaction occurs. It contains the cathode.

2.6 Half-cell

Half cell is a part of a cell, containing electrode dipped in an electrolytic solution. If oxidation occurs at the electrode that is called oxidation half cell. If reduction occurs at the electrode that is called reduction half cell.

2.7 Cell

Cell is a device consisting two half cell. The two half cells are connected through one wire. A cell is a device which converts electrical energy into chemical energy (or) chemical energy into electrical energy. Generally a cell consists of two half cells. Each half cell consists of an electrode dipped in an electrolytic solution. These two half cells are connected through one wire.

2.7.1 Type of Cell

Based on the type of reaction occurring in a cell, cells are classified into two types:

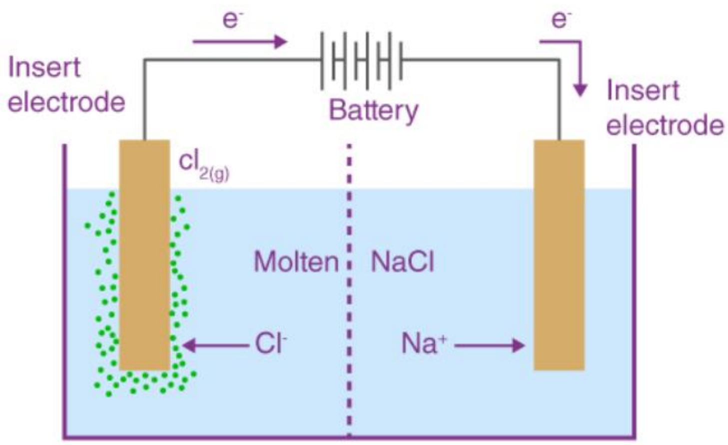
- **Electrolytic cells**
- **Electrochemical cells (or) voltaic cell (or) galvanic cells**

2.7.1.2 Electrolytic cells

Electrolytic cells are the cells in which electrical energy is used to bring about the chemical reaction. The three primary components of electrolytic cells are:

- Cathode (which is negatively charged for electrolytic cells)
- Anode (which is positively charged for electrolytic cells)
- Electrolyte

The electrolyte provides the medium for the exchange of electrons between the cathode and the anode. Commonly used electrolytes in electrolytic cells include water (containing dissolved ions) and molten sodium chloride. Molten sodium chloride (NaCl) can be subjected to electrolysis with the help of an electrolytic cell, as illustrated below.



Here, two inert electrodes are dipped into molten sodium chloride (which contains dissociated Na+ cations and Cl– anions). When an electric current is passed into the circuit, the cathode becomes rich in electrons and develops a negative charge. The positively charged sodium cations are now attracted towards the negatively charged cathode. This results in the formation of metallic sodium at the cathode.

Simultaneously, the chlorine atoms are attracted to the positively charged cathode. This results in the formation of chlorine gas (Cl_2) at the anode (which is accompanied by the liberation of 2 electrons, finishing the circuit). The associated chemical equations and the overall cell reaction are provided below.

- **Reaction at Cathode:** $[Na^+ + e^- \rightarrow Na] \times 2$
- **Reaction at Anode:** $2Cl^- \rightarrow Cl_2 + 2e^-$
- **Cell Reaction:** $2NaCl \rightarrow 2Na + Cl_2$

Thus, molten sodium chloride can be subjected to electrolysis in an electrolytic cell to generate metallic sodium and chlorine gas as the products.

2.7.1.3 Applications of Electrolytic Cells

- The primary application of electrolytic cells is for the production of oxygen gas and hydrogen gas from water.
- Another notable application of electrolytic cells is in electroplating, which is the process of forming a thin protective layer of a specific metal on the surface of another metal.
- It can be noted that the industrial production of high-purity copper, high-purity zinc, and high-purity aluminium is almost always done through electrolytic cells.

2.7.1.3 Galvanic cell or voltaic cell

An electrochemical cell that converts the chemical energy of spontaneous redox reactions into electrical energy is known as a galvanic cell or a voltaic cell. Example Daniel cell. Let us discuss this in detail in the next lesson plan.

Difference between electrochemical and electrolytic cells

S.No	Electrolytic cell	Electrochemical cell
1.	Electrical energy is converted to chemical energy	Chemical energy is converted to electrical energy
2.	The anode carries positive charge and cathode negative charge	The anode carries negative charge and cathode positive charge
3.	The electrons are supplied to the cell from the external battery	Electrons are drawn from the cell
4.	Amount of electricity passed during electrolysis is measured by coulometer	The e.m.f produced in the cell measured by potentiometer
5.	The extent of chemical reaction occurring at the electrodes is governed by Faradays law of electrolysis	The e.m.f of the cell depends upon the concentration of the electrolytes and the chemical nature of the electrode.
6.	Example: Electrolysis, electroplating, etc.	Example: Daniel cell

Discussion questions

1. Why does a galvanic cell produce electrical energy, while an electrolytic cell requires an external power source?
2. Assess the efficiency of different electrolytes in conducting electricity in an electrochemical cell.
3. How do the nature of the electrolyte and the electrodes affect the efficiency of an electrolytic process?
4. If you swap the electrodes of a galvanic cell, what happens to the direction of electron flow?
5. In an electrolysis experiment, how could you determine the amount of metal deposited on the cathode?